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Number Representations in LLMs: A Computational Parallel to Human Perception

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Abstract

Humans are believed to perceive numbers on a logarithmic mental number line, where smaller values are represented with greater resolution than larger ones. This cognitive bias, supported by neuroscience and behavioral studies, suggests that numerical magnitudes are processed in a sublinear fashion rather than on a uniform linear scale. Inspired by this hypothesis, we investigate whether large language models (LLMs) exhibit a similar logarithmic-like structure in their internal numerical representations. By analyzing how numerical values are encoded across different layers of LLMs, we apply dimensionality reduction techniques such as PCA and PLS followed by geometric regression to uncover latent structures in the learned embeddings. Our findings reveal that the model's numerical representations exhibit sublinear spacing, with distances between values aligning with a logarithmic scale. This suggests that LLMs, much like humans, may encode numbers in a compressed, non-uniform manner[1,2].

1 Introduction

Large language models (LLMs) have demonstrated impressive capabilities in natural language processing tasks (Touvron et al., 2023; Achiam et al., 2023), yet their internal representations of abstract concepts, i.e., numbers, space, and time, remain largely opaque. Recent research suggests that LLMs construct structured "world models," encoding relationships in ways that can be systematically analyzed (Petroni et al., 2019; Radford et al., 2019). For instance, studies have shown that spatial and geographical information is embedded in low-dimensional subspaces, where model performance correlates with data exposure (Gurnee and Tegmark, 2024; Godey et al., 2024). Similarly, numerical representation is influenced by tokenization strategies, with base-10 encoding proving more efficient for numeric reasoning tasks than higher-base tokenizations (Zhou et al., 2024).

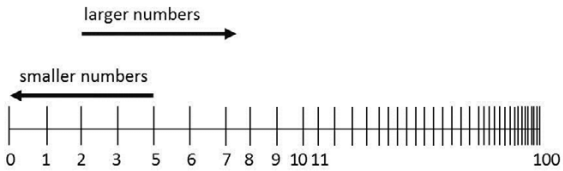

Figure 1: *Logarthmic mental number line hypothesis* asserts that humans innately percieve numbers on a logarithmic scale. Image source (Fritz et al., 2013).

The *linear hypothesis* of internal representations (Park et al., 2023) posits that concepts in LLMs are structured within geometric, linear subspaces, facilitating interpretability and manipulation. This framework suggests that numerical properties follow systematic, monotonic trends (Heinzerling and Inui, 2024). As a result, it has been commonly assumed that numerical values are represented in a uniform linear fashion (Zhu et al., 2025). However, recent probing studies (Zhu et al., 2025; Levy and Geva, 2024) challenge this assumption, revealing a non-uniform encoding of numbers in LLMs, where precision decreases for larger values. These findings raise questions about how artificial systems internalize numerical representations, particularly in relation to the scaling of numbers. Do LLMs preserve a uniform spacing of numerical values, and if not, what is the nature of their positioning?

Such questions naturally lead to an investigation of whether LLMs encode numerical values in a way that mirrors human cognition, as suggested by the *logarithmic mental number line hypothesis*. This hypothesis posits that humans perceive numerical magnitudes nonlinearly, following a logarithmic rather than a uniform linear scale (see Figure 1). Rooted in psychophysical studies like the Fechner-

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Weber law, this idea is supported by behavioral experiments showing that young children and individuals with limited formal education tend to map numbers logarithmically when placing them on a spatial axis (Fechner, 1860; Dehaene, 2003; Siegler and Opfer, 2003). While formal training shifts numerical perception toward a more linear scale, logarithmic encoding persists in tasks involving estimation and large-number processing (Dehaene et al., 2008; Moeller et al., 2009).

Inspired by this, we investigate whether LLMs encode numerical values in a manner analogous to the human logarithmic mental number line. By analyzing hidden representations across model layers, we examine the geometric structure of numerical magnitudes and their underlying trends. Our approach first employs dimensionality reduction techniques, including Principal Component Analysis (PCA) and Partial Least Squares (PLS), to transform the hidden representations onto a one-dimensional number line, that best fits its dominant numerical features. Second, using Spearman rank coefficient and geometric non-linear regression, we specifically test whether two key properties reminiscent of human numerical cognition (order preservation in representations and a compression effect where distances between consecutive numbers decrease as values increase) emerge in LLMs.

While both PCA and PLS reveal that numerical representations largely reside in a linear subspace, only PCA captures systematic sublinearity, suggesting that simple linear probes[3] may overlook the underlying non-uniformity in LLMs' numerical encoding.

Contributions We summarize our main finding in the following:

- We introduce a methodology for analyzing the geometric structure of number representations, offering a systematic approach to studying numerical abstractions in artificial neural networks.

- We provide empirical evidence that LLMs encode numerical values in a structured yet non-uniform way, revealing systematic compression reminiscent of the human logarithmic mental number line. Our findings refine the linear hypothesis by showing that numerical magnitudes in LLMs are not evenly spaced but follow a structured compression pattern.

[3]PLS is a linear probe that projects input data onto a lower-dimensional subspace, maximizing covariance with the target.

2 Related Works

Linearity of internal representations (Park et al., 2023) has been a central assumption in existing research, suggesting that language models encode numerical values in a linear manner. However, Zhu et al. (2025) present a more nuanced perspective. Their analysis of partial number encoding (Appendix F) shows that probing accuracy declines as sequence length increases, with greater difficulty in capturing precise values at larger scales, a pattern reminiscent of logarithmic encoding, where resolution is higher for smaller numbers. Some of the conclusions in Zhu et al. (2025) are that LLMs encode numerical values in their hidden representations, yet linear probes fail to precisely reconstruct these values, as discussed in Zhu et al. (2025, Section 3.1). The authors there suggest that "This phenomenon may indicate that language models use stronger non-linear encoding systems". Our findings support this claim and further uncover the underlying nature of this non-linearity.

Recent studies such as Levy and Geva (2024); Zhou et al. (2024) show that LLMs rely on base-10 digit-wise representations rather than encoding numbers in a continuous linear space, as revealed through circular probing techniques. While individual digits are accurately reconstructed, performance declines for larger numbers, suggesting a structured rather than holistic encoding. Furthermore, Zhou et al. (2024) demonstrate that LLMs trained on higher-base numeral systems struggle with numerical extrapolation, implying an implicit compressed representation where smaller values have finer granularity—consistent with logarithmic scaling. Collectively, these results align with and further substantiate the hypothesis that LLMs internally represent numbers in a non-uniform, sublinear manner.

Logarithmic functions can appear linear over small local intervals, which may explain why LLMs are often assumed to represent numbers linearly. Consequently, methods like PLS regression and activation patching (Heinzerling and Inui, 2024; El-Shangiti et al., 2024), which analyze small activation variations, may capture local monotonicity while missing the global nonlinear structure. This suggests that reported linear effects could stem from analyzing narrow numerical ranges, whereas

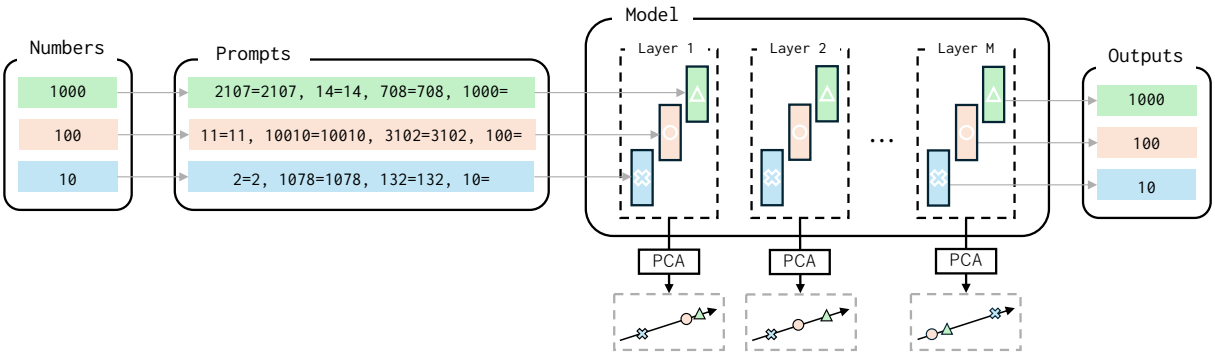

Figure 2: The overall graphical representation of our method. Numbers are passed to the model in form of a prompt and the internal representations are captured from the embeddings corresponding to token '='. At every layer, we perform PCA projections onto one and two dimensional subspaces and pick a layer with highest explained variance (σ^2) score to further analyze monotonicity and scaling of number representations.

a broader examination may reveal an underlying logarithmic representation.

Finally, we also emphasize the difference between our work and prior studies on numerical reasoning (Park et al., 2022; Zhang et al., 2020) which evaluate models' ability to process explicit numbers rather than probing their internal representations. While Park et al. (2022) focus on tasks like unit conversion and range detection, Zhang et al. (2020) examine numerical magnitude in common sense reasoning. Unlike these works, our study investigates the spatial structure of numerical representations within hidden states and how this encoding generalizes across scales.

3 Methodology

3.1 General settup

The *logarithmic mental number line hypothesis* (Dehaene et al., 2008) suggests that humans innately perceive numerical magnitudes on a logarithmic scale rather than a linear one. Formally, if we denote the internal mapping of numbers to their cognitive representation as f_{H}, the hypothesis asserts that f_{H} is approximately logarithmic. While the exact nature of f_{H} remains elusive, we adopt the guiding principle $f_{\text{H}} \equiv \log$ in our experiments.

On the other side, Large language models, like LLaMA-2, process inputs by mapping them into a high-dimensional representation space, where each input x (e.g., a number) is transformed into an internal representation $f(x) \in \mathbb{R}^d$. Analyzing the geometry of these representations across a set of inputs $\mathcal{X}$ can reveal how the model organizes and reasons about them, shedding light on emergent properties such as whether the model encodes numerical values along a number line or exhibits cognitive-like patterns, such as sublinear scaling.

Our goal is **to study the properties of the function f_{LLM}, which serves as a counterpart of the human cognitive mapping f_{H} described above**. Specifically, we examine whether f_{LLM} preserves the natural ordering of numbers and how it transforms their magnitudes.

3.2 Definition of f_{LLM}

To analyze the structure of hidden representations, we apply a projection $T : \mathbb{R}^d \to \mathbb{R}^p$, $p = 1, 2$, obtained from techniques such as Principal Component Analysis (PCA) or Partial Least Squares (PLS). Then, our function is given by

$$f_{\text{LLM}}(x) := T(f(x)), \tag{1}$$

where f is a map between the input number and the corresponding internal representation in the model (see Sections 4 and 5 how f is defined in various settings).

For any two inputs $x, y \in \mathcal{X}$, we define the distance between their projections following the mapping T using Euclidean norm as

$$d(x, y) = ||f_{\text{LLM}}(x) - f_{\text{LLM}}(y)||. \tag{2}$$

3.3 Abstract Number Line and Monotonicity Metric

If the projection dimension is $p = 1$, one-dimensional embedding of numerical inputs forms a *number line* if the projections preserve monotonicity (resp. reverse monotonicity), i.e., for $x_1 < x_2$ (resp. $x_1 > x_2$), we have $f_{\text{LLM}}(x_1) < f_{\text{LLM}}(x_2)$. This ensures that the natural order of numerical values is maintained in the representation space.